Text Embeddings

Introduction

Embeddings are dense numerical vector representations of data.

Instead of storing text as words, embeddings convert text into vectors that capture semantic meaning.

Example:

"dog" → [0.24, -0.61, 0.87, ...]

"puppy" → [0.26, -0.58, 0.84, ...]

Since these vectors are similar, AI understands that both words have related meanings.

Why Embeddings?

Computers cannot understand natural language directly.

Embeddings convert language into mathematical space where similar concepts are located close together.

Types of Embeddings

Word Embeddings

Examples:

- Word2Vec
- GloVe
- FastText

Each word has its own vector.

Sentence Embeddings

Represent an entire sentence.

Examples:

- Sentence-BERT
- Universal Sentence Encoder

Document Embeddings

Represent complete documents for semantic search and retrieval.

Similarity Measurement

The most common metric is Cosine Similarity.

Cosine Similarity ranges from:

- 1 → identical meaning
- 0 → unrelated
- -1 → opposite

Formula:

Cosine(A,B)=A·B / ||A||||B||

Embeddings in RAG Systems

Workflow:

Documents

↓

Chunking

↓

Embedding Model

↓

Vector Database

↓

Similarity Search

↓

LLM

When a user asks a question, the query is embedded and matched against stored vectors.

Popular Embedding Models

- OpenAI text-embedding-3-large
- BAAI BGE
- E5
- Instructor XL
- all-MiniLM-L6-v2

Applications

- Semantic Search
- Recommendation Systems
- Question Answering
- Retrieval-Augmented Generation
- Duplicate Detection
- Clustering
- Document Search

Conclusion

Embeddings enable AI systems to understand semantic meaning rather than relying on exact keyword matching.